



Sri Chaitanya IIT Academy.,India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant**ICON Central Office - Madhapur - Hyderabad**

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 21-12-2025

Time: 09.00Am to 12.00Pm

GTM-03

Max. Marks: 300

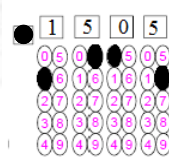
IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions**. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics and Chemistry** having **25 Questions** in each subject and subject having **two sections**.
(I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
(II) Section-II contains **05 Numerical Value Type Questions**.

- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted Question 21. Bubble on 21 as shown below

**Question Answered for Marking** **Question Cancelled for Marking****Marking scheme: +4** for correct answer, **0** if not attempt and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

--	--	--	--	--	--	--	--	--

Candidate's Signature: _____

Invigilator's Signature: _____

Sec: Sr. XL-500, Sr.S60(N&S) & GLIIT

Page 1

**Sri Chaitanya**
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1**
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**JEE MAIN****2023**

SINGARAJU

VENKAT KOUNDINIA

Sri Chaitanya

All India Rank 1

300

RANK**1****JEE Advanced****2023**

VAVILALA

CHIDVILAS REDDY

Sri Chaitanya

All India Rank 1

341

RANK**1****NEET****2023**

BORA VARUN

CHAKRAVARTHI

Sri Chaitanya

All India Rank 1

278

RANK**1****For More Join: @IITJEE_Advanced**



MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

Sec: **Sr. XL-500, Sr.S60(N&S) & GLIIT**

Page 2



Sri Chaitanya
Educational Institutions

Infinity
Learn



**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**

SINGARAJU
VENKAT KOUNDINYA
Sri Chaitanya
JEE 12th Class

**306
100**

RANK

1

**JEE Advanced
2023**

VAVILALA
CHIDVILAS REDDY
Sri Chaitanya
JEE 12th Class

**341
100**

RANK

1

**NEET
2023**

BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
JEE 12th Class

**700
100**

RANK

1

For More Join: @IITJEE_Advanced

**MATHEMATICS****Max Marks: 100****SECTION-I****(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- Two cards are drawn at random without replacement from a well-shuffled deck of 52 cards. Let X denote the random variable of number of face cards obtained in two drawn cards, then $P(1 \leq X \leq 2)$ is equal to $\frac{p}{q}$ (where p and q are co-prime natural numbers).
Then the value of $p + q$ is
1) 13 2) 18 3) 24 4) 12
- For $a \in \mathbb{C}$ let $A = \{z \in \mathbb{C} : \operatorname{Re}(a + \bar{z}) > \operatorname{Im}(\bar{a} + z)\}$ and $B = \{z \in \mathbb{C} : \operatorname{Re}(a + \bar{z}) < \operatorname{Im}(\bar{a} + z)\}$.
Then among the two statements:
S1: If $\operatorname{Re}(a), \operatorname{Im}(a) > 0$, then the set A contains all the real numbers
S2: If $\operatorname{Re}(a), \operatorname{Im}(a) < 0$, then the set B contains all the real numbers
1) Only (S1) is true 2) both are false
3) Only (S2) is true 4) Both are true
- $\lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{\sum_{r=1}^{3n} \frac{r}{n^2 \sqrt{16n^2 + r^2}}}$ equals
1) e 2) $\frac{1}{e^3}$ 3) -1 4) $\frac{1}{e}$
- Let f and g be continuous and differentiable functions. If $f(0) = f(2) = f(4)$;
 $f(1) + f(3) = 0$; $g(0) = g(2) = g(4) = 0$ and if $f(x) = 0$ and $g'(x) = 0$ do not have a common root, then the minimum number of zeroes of $f'(x)g'(x) + f(x)g''(x)$ in $[0, 4]$ is
1) 3 2) 4 3) 5 4) 2

Sec: **Sr. XL-500, Sr.S60(N&S) & GLIIT**

Page 3

**Sri Chaitanya**
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023****JEE MAIN
2023**SINGARAJU
VENKAT KOUNDINYA
Sri Chaitanya
JEE Main Rank 1
30th Class**30th**
Rank**RANK****1****JEE Advanced
2023**VAVILALA
CHIDVILAS REDDY
Sri Chaitanya
JEE Advanced Rank 1
34th Class**34th**
Rank**RANK****1****NEET
2023**BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
NEET Rank 1
28th Class**28th**
Rank**RANK****1****For More Join: @IITJEE_Advanced**



5. If A is a square matrix of order 2, and $(tr(A))^2 - tr(A^2) = k \det(A)$, where k lies between the roots of equation $x^2 - ax + (a-1) = 0$, then least integral value of a is
 1) 3 2) 4 3) 5 4) 2
6. The director circle of a hyperbola is $x^2 + y^2 - 6x - 8y + 5 = 0$. If one end of the transverse axis of hyperbola is $(3, -1)$, then one of the focus of this hyperbola is
 1) $(3 + 5\sqrt{6}, 4)$ 2) $(3 + 5\sqrt{30}, 4)$ 3) $(3, 4 + \sqrt{30})$ 4) $(3, 4)$
7. The sum of the series $8! + \frac{9!}{2!} + \frac{10!}{3!} + \frac{11!}{4!} + \dots$ upto 20 terms is
 1) $7!(^{27}C_{19} - 1)$ 2) $8!(^{27}C_{20} - 1)$ 3) $7!(^{28}C_{20} - 1)$ 4) $8!(^{28}C_{20} - 1)$
8. If the chords of the hyperbola $x^2 - y^2 = 16$ touches the parabola $y^2 = 16x$, then the locus of the middle points of these chords satisfies the equation
 1) $y^2(x+4) = x^3$ 2) $y^2(x-4) = x^3$
 3) $y^2(x+8) = 3x^3$ 4) $y^2(x-8) = 2x^3$
9. Let $S = \{(x, y) : x^2 + y^2 - 4x - 2y + 4 = 0 \text{ \& } x^2 + y^2 - 2y = 0\}$, τ = set of circles which pass through all point(s) in S. Then the number of elements in τ with unit radius that pass through all the point(s) in S and the origin is
 1) 0 2) 1 3) 2 4) 3
10. Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k + x\sqrt{1-y^2}$, where k is a constant and $y\left(\frac{1}{2}\right) = \frac{1}{4}$. Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$ is equal to
 1) $\frac{2}{\sqrt{5}}$ 2) $-\frac{\sqrt{5}}{4}$ 3) $-\frac{\sqrt{5}}{2}$ 4) $\frac{\sqrt{5}}{2}$





11. If the area of the region bounded by the curves $xy + x = 1$, $y^2 = 12x$ and the x-axis is A, then
 1) $A > 1$ 2) $A < \ln 2$ 3) $A^2 < \frac{1}{\sqrt{2}}$ 4) $A > \frac{1}{\sqrt{2}}$
12. 5 boys (B_1, B_2, B_3, B_4 and B_5) and 5 girls (G_1, G_2, G_3, G_4 and G_5) are to be seated around a round table such that boy and girl sit alternately and B_i does not sit beside $G_i \forall i \in \{1, 2, 3, 4, 5\}$. If the number of such arrangements is N, then the sum of digits of N is equal to
 1) 8 2) 7 3) 6 4) 5
13. If $x = \int_0^t e^{\sqrt{z}} \left\{ \frac{2 \tan \sqrt{z} + 1 - \tan^2 \sqrt{z}}{2\sqrt{z} \sec^2 \sqrt{z}} \right\} dz$ & $y = \int_0^t e^{\sqrt{z}} \left\{ \frac{1 - \tan^2 \sqrt{z} - 2 \tan \sqrt{z}}{2\sqrt{z} \sec^2 \sqrt{z}} \right\} dz$.
 Then the inclination of the tangent to the curve at $t = \frac{\pi}{4}$ with positive x axis is
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\frac{3\pi}{4}$
- 14.

Column-I		Column-II	
A)	If range of $f(x) = \cos^{-1} \frac{x}{2} + x^2 - 4x + 1$ is $[a, \pi + b]$, then $a + b$ is equal to	P)	1
B)	Length of the tangent drawn from a point on the circle $x^2 + y^2 = 25$ to the circle $x^2 + y^2 = 21$ is	Q)	2
C)	If $\int_0^6 e^{3x - [3x]} dx = p(e - q)$, where $[.]$ denotes greatest integer function, the $p + q$ is.....where $p, q \in N$	R)	7
D)	Minimum possible number of positive roots of the quadratic equation $x^2 - (1 + b)x + b - 2 = 0, b \in R$ is	S)	10

- 1) $A \rightarrow S, B \rightarrow Q, C \rightarrow R, D \rightarrow P$ 2) $A \rightarrow P, B \rightarrow Q, C \rightarrow R, D \rightarrow S$
 3) $A \rightarrow S, B \rightarrow Q, C \rightarrow R, D \rightarrow R$ 4) $A \rightarrow S, B \rightarrow P, C \rightarrow R, D \rightarrow P$





15. If 6th term of G.P is 96 & n^{th} term is lying between 500 & 780, where first term is a & ($[a] \neq 0$, where $[.]$ denotes greatest integer function) & common ratio is positive integer ($r \neq 1$), then number of terms in the series are....(consider n^{th} term is the last term of series)
- 1) 8 2) 9 3) 10 4) 11
16. Let $\vec{a} = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix}$ & $\vec{c} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$. If V_1 is the volume of parallelopiped whose three coterminous edges are the vectors $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$, $\vec{c} + \vec{a}$ & V_2 is the volume of tetrahedron whose coterminous edges are the vectors $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{c}$, $\vec{c} \times \vec{a}$, then the value of $(V_1 + V_2)$ is
- 1) 256 2) 278 3) 288 4) 289
17. Let $L_1: \frac{x-6}{3} = \frac{y-7}{-1} = \frac{z-4}{1}$ and $L_2: \frac{x}{-3} = \frac{y+9}{2} = \frac{z-2}{4}$. If L_1 & L_2 are skew lines such that P & Q are two points nearest to each other lying on lines L_1 & L_2 respectively. If image of P with respect to plane $3x + 3y - z = 11$ is $R(a, b, c)$. Then $(a + b + c)$ equals
- 1) 4 2) 3 3) 5 4) 1
18. If $3^{\sin 2x + 2 \cos^2 x} + 3^{1 - \sin 2x + 2 \sin^2 x} = 28$, then x is
- 1) $\frac{(4k-1)\pi}{4}, (2k+1)\frac{\pi}{2}, k \in \mathbb{Z}$ 2) $\frac{(2k\pi + (-1)^k \pi)}{4}, k\pi, k \in \mathbb{Z}$
- 3) ϕ 4) $\frac{k\pi}{2}, \frac{(3k+1)\pi}{3}, k \in \mathbb{Z}$
19. Range of $y = \cos\left(\pi \sin\left(\frac{\pi}{2} \cos(\pi \sin x)\right)\right)$, where $x \in \mathbb{R}$, is
- 1) $[-1, 1]$ 2) $[-\pi, \pi]$ 3) $[0, 1]$ 4) $[-1, 0]$
20. Let I be the set of positive integers. R is a relation on the set I given by
- $$R = \left\{ (a, b) \in I \times I : \log_2 \left(\frac{a}{b} \right) \text{ is a non-negative integer} \right\}$$
- , then R is
- 1) neither symmetric nor transitive but reflexive.
 2) reflexive, transitive but not symmetric
 3) neither reflexive nor transitive but symmetric
 4) equivalence relation





SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Given the system of equation

$$x + y - z = 2,$$

$$2x - y + 4z = 1 \text{ and}$$

$$x + \alpha y + z = \beta, \text{ where } \alpha, \beta \in \{0, 1, 2, 3, 4\}$$

Let A, B, C denotes the number of ordered pairs (α, β) such that the system has unique solution, no solution, infinitely many solutions respectively, then the value of $(A - 4B + C)$ is equal to

22. Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix}; a, b \in \{1, 2, 3, \dots, 100\} \right\}$ and let $T_n = \left\{ A \in S : A^{n(n+1)} = I \right\}$. Then the

number of elements in $\bigcap_{n=1}^{100} T_n$ is ____.

23. Let $x = 2$ be a root of the equation $x^2 + px + q = 0$.

$$\text{and } f(x) = \begin{cases} \frac{1 - \cos(x^2 - 4px + q^2 + 8q + 16)}{(x - 2p)^4}, & x \neq 2p \\ 0, & x = 2p \end{cases}$$

Then $\lim_{x \rightarrow 2p^+} [f(x)]$ is....

Where $[.]$ denotes greatest integer function,

24. For a data of n observations $x_1, x_2, \dots, x_n$, if $\sum_{i=1}^n (2x_i + 1)^2 = 29n$ and $\sum_{i=1}^n (3x_i - 1)^2 = 34n$, then the variance of this data is

25. The number of non differentiable point(s) of function $f(x) = |x^2 - 2x - 3| \cdot e^{|9x^2 - 12x + 4|}$, $x \in R$ is



JEE MAIN

2023

SINGARAJU
VENKAT KOUNDINYA
Sri Chaitanya
JEE Main Rank 1

306

RANK

RANK

1

JEE Advanced

2023

VAVILALA
CHIDVILAS REDDY
Sri Chaitanya
JEE Advanced Rank 1

341

RANK

RANK

1

NEET

2023

BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
NEET Rank 1

206

RANK

RANK

1



PHYSICS

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

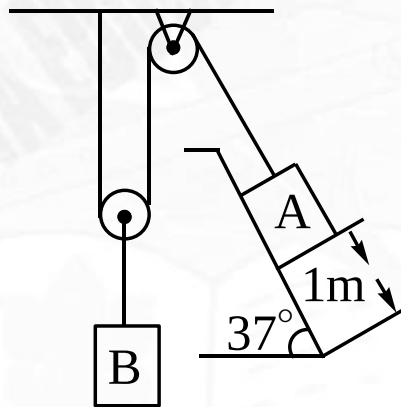
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. A particle starts from the origin of coordinate system at $t = 0$ and moves in the XY plane with a constant acceleration α in the Y direction. Its equation of motion is

$y = \beta x^2$. Then its velocity component in the X direction is

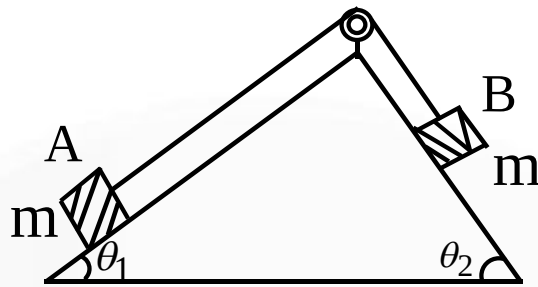
- 1) Zero 2) $\sqrt{\frac{2\alpha}{\beta}}$ 3) $\frac{\alpha}{2\beta}$ 4) $\sqrt{\frac{\alpha}{2\beta}}$

27. The block A and B shown in the figure have masses $M_A = 5\text{ kg}$ and $M_B = 4\text{ kg}$. The system is released from rest. The speed of B after A has travelled a distance 1 m along the incline is



- 1) $\frac{2\sqrt{g}}{\sqrt{3}}$ 2) $\frac{\sqrt{3}}{4}\sqrt{g}$ 3) $\frac{\sqrt{g}}{2\sqrt{3}}$ 4) $\frac{\sqrt{g}}{\sqrt{3}}$
28. A body A of mass m slides on plane inclined at angle θ_1 to the horizontal and μ is the coefficient of friction between A and the plane. A is connected by a light string passing over a frictionless pulley to another body B also of mass m on sliding on a frictionless plane inclined at angle θ_2 to the horizontal. The correct statement among the following is





1) A will never move up the plane.

2) A will just start moving up the plane when $\mu = \frac{\sin \theta_2 - \sin \theta_1}{\cos \theta_1}$.

3) For A to move up the plane, θ_2 must always be lesser than θ_1 .

4) B will always slide down with constant speed.

29. The work done on a particle of mass m by a force,

$$\vec{F} = K \left[\frac{x}{(x^2 + y^2)^{3/2}} \hat{i} + \frac{y}{(x^2 + y^2)^{3/2}} \hat{j} \right],$$

K being the constant of appropriate dimensions, when the particle is taken from the point $(a, 0)$ to the point $(0, a)$ along a circular path of radius a about the origin in the x - y plane is

1) $\frac{2Kx}{a}$

2) $\frac{Kx}{a}$

3) $\frac{Kx}{2a}$

4) 0

30. A small sphere of mass 0.2 kg is attached to a light string of length 1.3 m whose upper end is fixed to a ceiling. The sphere is made to describe a horizontal circle of radius 0.5 m. Then the time taken to complete one revolution is:

1) $\frac{2\pi}{5} \sqrt{3} \text{ sec}$

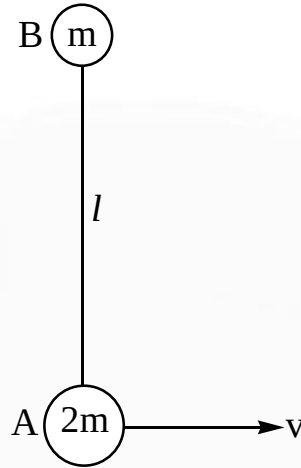
2) $\frac{\pi}{5} \sqrt{3} \text{ sec}$

3) $\frac{\pi}{5} \text{ sec}$

4) $\frac{2\pi}{5} \text{ sec}$

31. Two masses A and B connected with an inextensible string of length l lie on a smooth horizontal plane. A is given a velocity of v m/s along the ground perpendicular to line AB as shown in Fig. Find the tension in string during their subsequent motion.

**JEE MAIN
2023**SINGARAJU
VENKAT KOUNDINYA
AIR 1st Rank
Sri Chaitanya
JEE 12th Class**306**
RANK**RANK****JEE Advanced
2023**VAVILALA
CHIDVILAS REDDY
AIR 341st Rank
Sri Chaitanya
JEE 12th Class**341**
RANK**RANK****NEET
2023**BORA VARUN
CHAKRAVARTHI
AIR 2nd Rank
Sri Chaitanya
JEE 12th Class**2nd**
RANK**RANK****1**



- 1) $\frac{2mv^2}{3l}$ 2) $\frac{3mv^2}{2l}$ 3) $\frac{2mv^2}{l}$ 4) $\frac{mv^2}{3l}$

32. A uniform rod AB of mass m and length l is at rest on a smooth horizontal surface. An impulse 'J' is applied to the end 'B' perpendicular to the rod in horizontal direction.

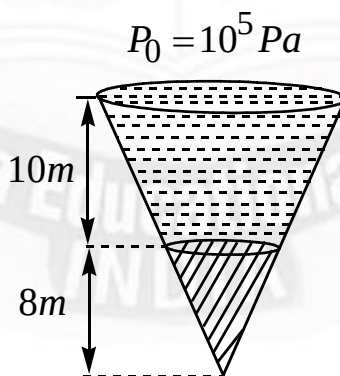
Speed of particle P at distance $\frac{l}{6}$ from the centre towards A of the rod after time

$t = \frac{\pi ml}{12J}$ is:

- 1) $2\frac{J}{m}$ 2) $\frac{J}{\sqrt{2}m}$ 3) $\frac{J}{m}$ 4) $\sqrt{2}\frac{J}{m}$

33. The figure shows a conical container of half-apex angle 37° filled with certain amount of kerosene and water as shown. Specific gravity of kerosene is 0.8 and axis of cone is vertical. The force exerted by the water on the kerosene in approximately.

(Take atmospheric pressure $= 10^5 \text{ Pa}$ and $g = 10 \text{ m/s}^2$)

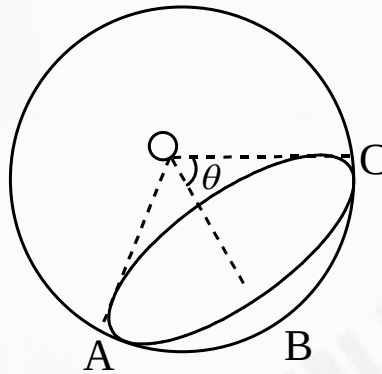


- 1) $3 \times 10^6 \text{ N}$ 2) $4 \times 10^7 \text{ N}$ 3) $2 \times 10^7 \text{ N}$ 4) $5 \times 10^7 \text{ N}$

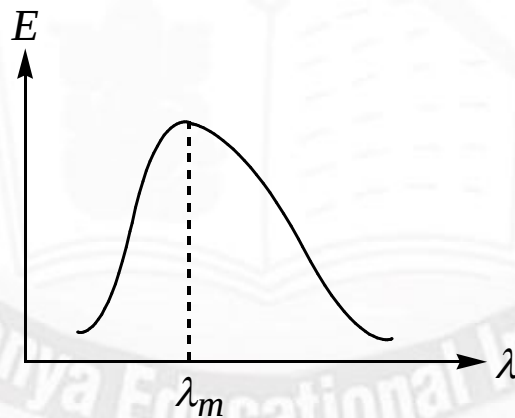




34. Consider a small water drop in air. Let T be the surface tension of water. The figure shows a small section ABC of the drop subtending angle 2θ at the centre O. The force due to surface tension acting on this section is (R = radius of the drop)



- 1) $2\pi TR \sin^3 \theta$ 2) $4\pi TR \sin \theta$ 3) $2\pi TR \sin^2 \theta$ 4) $2\pi TR \sin \theta$
35. The radiations by the sun are analysed and the spectral energy distribution curve is plotted, as shown. The radius of the sun is R . Treat the sun as perfectly black body which radiate power at constant rate till its store of hydrogen gets exhausted. Assume the sun produces energy only by nuclear fusion of hydrogen nuclei and the fraction of nuclear mass of hydrogen that can be converted to energy is η . How long will the sun keep on emitting the energy if the mass of hydrogen at present in Sun's core is m ? (Stefan's constant = σ , Wien's constant = b , speed of light = c)

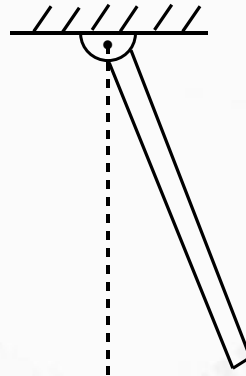


- 1) $\frac{\eta mc^2 \lambda_m^4}{\pi R^2 \sigma b^4}$ 2) $\frac{\eta mc^2 \lambda_m^4}{4\pi R^2 \sigma b^4}$ 3) $\frac{\eta mc^2 \lambda_m^4}{2\pi R^2 \sigma b^4}$ 4) $\frac{2\eta mc^2 \lambda_m^4}{\pi R^2 \sigma b^4}$

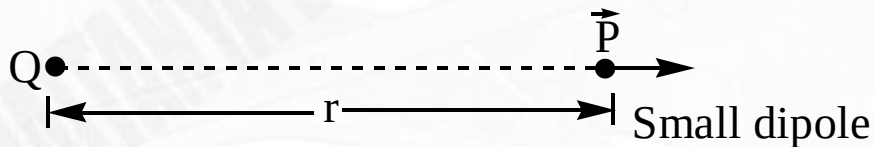
**JEE MAIN**
2023SINGARAJU
VENKAT KOUNDINYA
4th Rank
Sri Chaitanya
6th 12th Class**306**
RANK**RANK****1****JEE Advanced**
2023VAVILALA
CHIDVILAS REDDY
11th Rank
Sri Chaitanya
6th 12th Class**341**
RANK**RANK****1****NEET**
2023BORA VARUN
CHAKRAVARTHI
11th Rank
Sri Chaitanya
6th 12th Class**206**
RANK**RANK****1**



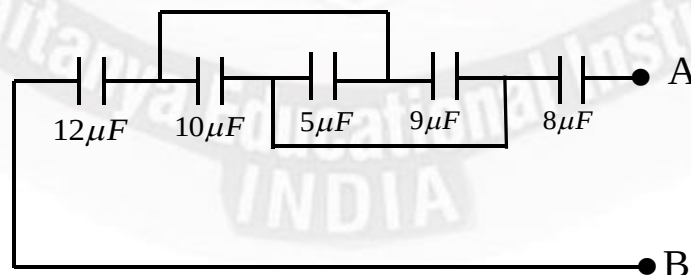
36. A meter stick swinging in vertical plane about a fixed horizontal axis passing through its one end undergoes small oscillations of frequency f_0 . If the bottom half of the stick is cut off, then its new frequency of small oscillation would become:



- 1) $\frac{f_0}{\sqrt{2}}$ 2) $\sqrt{2} f_0$ 3) f_0 4) $2\sqrt{2} f_0$
37. For the situation shown in the figure below, find the magnitude of force experience by the dipole, due to charge Q place at distance r from the dipole $\vec{P}$.



- 1) $\frac{|\vec{P}|Q}{2\pi\epsilon_0 r^3}$ 2) $\frac{|\vec{P}|Q}{2\pi\epsilon_0 r^2}$ 3) $\frac{3Q|\vec{P}|}{2\pi\epsilon_0 r^3}$ 4) 0
38. Five capacitors together with their capacitances are shown in the adjoining figure. The potential difference between the point A and B is 60 volt. The equivalent capacitance between the point A and B and charge on capacitor $5\mu F$ will be respectively are:

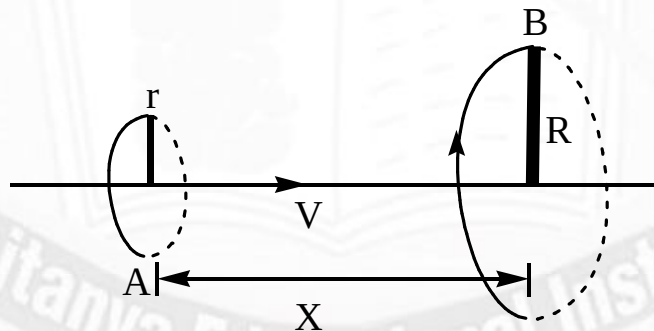


- 1) $44\mu F, 30\mu C$ 2) $4\mu F, 150\mu C$ 3) $15\mu F, 200\mu C$ 4) $4\mu F, 50\mu C$





39. A charged particle of charge q , mass m when projected perpendicular to a uniform and constant magnetic field was found to execute a circular motion of radius R . When the same particle is projected at angle $\theta (\neq 0^\circ, 90^\circ, 180^\circ)$ to the same magnetic field and at same speed as before it executes a curvilinear motion of radius of curvature r . The value of r is
- 1) $R \cos^2 \theta$ 2) $R \cot^2 \theta$ 3) $R \operatorname{cosec} \theta$ 4) $R \cot \theta$
40. A compass needle is placed in the gap of a parallel plate capacitor. The capacitor is connected to a battery through a resistance. The compass needle
- 1) Does not deflect
2) Deflects for a very short time and then comes back to original position
3) Deflects and remains deflected as long as the battery is connected
4) Deflects and gradually comes to the original position in a time which is large compared to the time constant.
41. Loop A of radius ($r \ll R$) moves towards loop B carrying some current, with a constant velocity V in such a way that their planes are always parallel. What is the distance between the two loops (x) when the induced emf in loop A is maximum



- 1) R 2) $\frac{R}{\sqrt{2}}$ 3) $\frac{R}{2}$ 4) $R \left(1 - \frac{1}{\sqrt{2}} \right)$

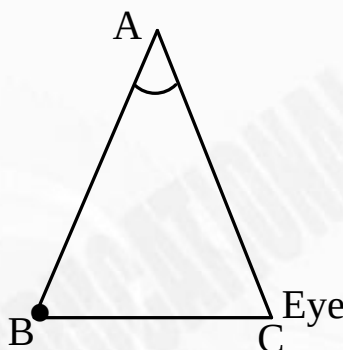




42. In an ideal transformer, the voltage and the current in the primary coil are 200V and 2A respectively. If the voltage in the secondary coil is 2000V, then the value of current in the secondary coil will be

- 1) 0.2A 2) 2A 3) 10A 4) 20A

43. A black spot B is marked very near to base of the prism ($\mu = 3/2$) as shown in figure. If the image of black spot appears on the surface AC of the prism, the angle of the prism A will be



- 1) $A < \sin^{-1}\left(\frac{2}{3}\right)$ 2) $A = \sin^{-1}\left(\frac{2}{3}\right)$
 3) $\sin^{-1}\left(\frac{2}{3}\right) < A < 2\sin^{-1}\left(\frac{2}{3}\right)$ 4) $A = 2\sin^{-1}\left(\frac{2}{3}\right)$

44. Given below are two Statements: One is labelled as Assertion (A) and the other labelled as Reason(R).

Assertion A: The binding energy per nucleon is practically independent of the atomic number for nuclei of mass number in the range 30 to 170.

Reason R: Nuclear force is short ranged.

In the light of the above Statements, choose the correct answer from the options given below.

- 1) A is true but R is false
 2) Both A and R are true but R is not the correct explanation of A.
 3) A is false but R is true
 4) Both A and R are true and R is the correct explanation of A





45. Assertion (A): The resistance of an intrinsic semiconductor decreases with increases in its temperature.

Reason: The number of conduction electrons as well as hole increases in an intrinsic semiconductor with rise in its temperature.

1) Both Assertion (A) Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

3) Assertion (A) is true, but Reason (R) is false

4) Assertion (A) is false, but Reason (R) is true

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. Electromagnetic radiation whose electric component varies with time as $E = C_1(C_2 + C_3 \cos \omega t) \cos \omega_0 t$, here C_1, C_2 and C_3 are constants, is incident on lithium and liberates photoelectrons. If the kinetic energy of most energetic electrons be 2.6 eV, the work function of lithium is (in eV). [Take: $\omega_0 = 2.4\pi \times 10^{15} \text{ rad/sec}$ and

$\omega = 8\pi \times 10^{14} \text{ rad/sec}$, planks constant $h = 6.6 \times 10^{-34}$ in SI units]

47. A hydrogen-like atom (atomic number Z) is in a higher excited state of quantum number n . This excited atom can make a transition to the first excited state by successively emitting two photons of energies 10.20 eV and 17.00 eV respectively. Alternatively the same atom from the same excited state can make a transition to second excited state by successively emitting two photons of energies 4.25 eV and 5.95 eV respectively.

Determine the value

of n (Ionization energy of hydrogen atom = 13.6 eV).

48. The diameter of a wire of length 50cm is measured with the help of a screw gauge. The main scale reading is 1mm and circular scale reading is 25. Pitch of the screw gauge is 1mm and the total number of divisions on the circular scale is 100. This wire is used in an experiment of determination for the Young's modulus of material of a wire by Searle's method. The following data are available: elongation in the wire $l = 0.125\text{cm}$ under the tension of 50N, least count for measuring normal length of wire is 0.01cm and for elongation in the wire is 0.001cm. The maximum percentage error in calculating value of

JEE MAIN
2023SINGARAJU
VENKAT KOUNDINYA
Sri Chaitanya
JEE 12th Class306
RANK

RANK

JEE Advanced
2023VAVILALA
CHIDVILAS REDDY
Sri Chaitanya
JEE 12th Class341
RANK

RANK

NEET
2023BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
JEE 12th Class278
RANK

RANK

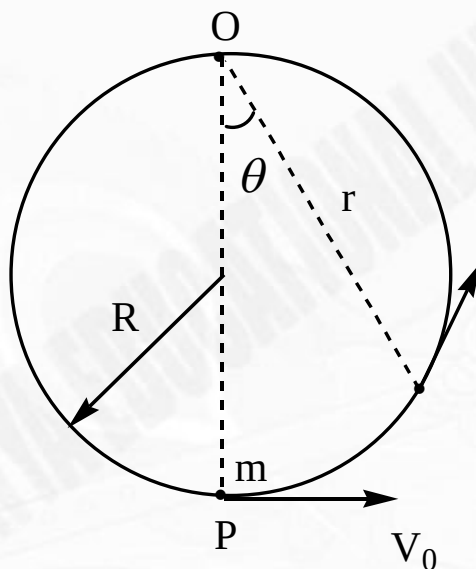


Young's modulus (Y) is $n \times 0.484$. Find n . (assuming that the force is measured very accurately, $Y = Fl / A\Delta l$, where symbols have their usual meaning).

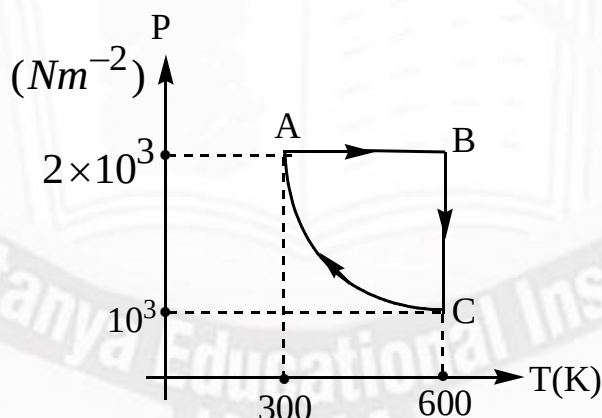
49. A particle is moving along a circle of radius R . Particle is moving under the Influence of an unknown gravitational central force F which is always pointing towards point O . particle is projected from point P which is directly opposite of O with a initial velocity V_0

as shown in the figure. If the expression of force F is given by $F = \frac{32m^x R^y V_0^z}{r^K}$,

Where x, y, z and K are integers. Then the value of $\{(x + y + z) - K\}$ will be?



50. Two moles of a monatomic ideal gas is taken through a cyclic process as shown.



Process CA follows $PT = \text{constant}$. If cycle efficiency is $1 - \frac{21}{12 \ln 2 + 3x}$, then find the value of x ?

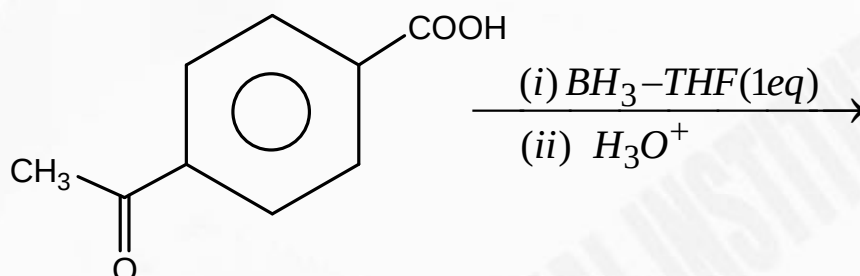


**CHEMISTRY****Max Marks: 100****SECTION-I****(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

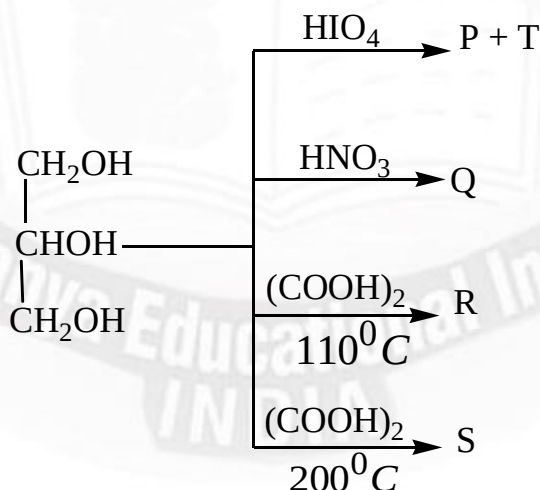
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. The major product obtained in the following reaction is



- 1)
- 2)
- 3)
- 4)

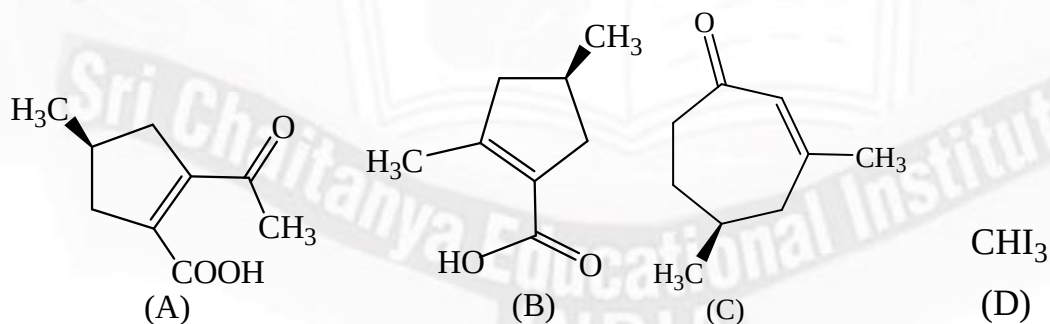
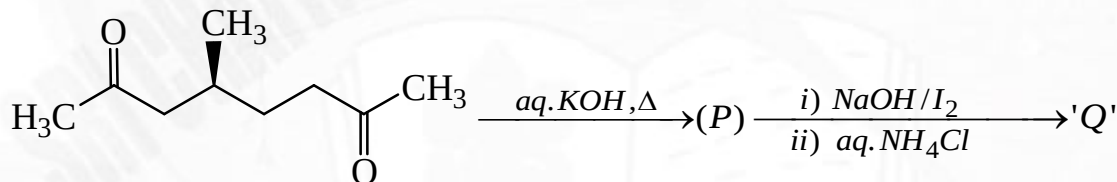
52. The correctly matched option with unknown products in the following is





P	Q	R	S	T
1) $CH_2 = O$		HCOOH		HCOOH
2) HCOOH		CH_3COOH		$CH_2 = O$
3) HCOOH				
4) $CH_2 = O$			HCOOH	HCOOH

53. The major product 'Q' in the following sequence of reaction is



- 1) (A) 2) (B) 3) (C) 4) (D)

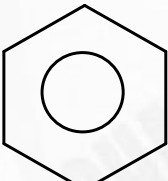
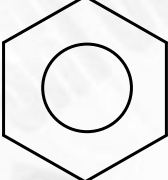
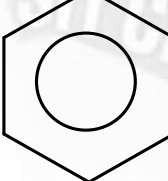
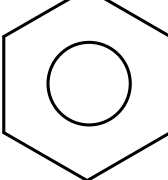
54. Which of the following is an essential amino acid with two chiral center

- 1) Histidine 2) Tryptophan 3) Lysine 4) Isoleucine





55. Which of the following is incorrect regarding hormones
- 1) Hormones acts as intercellular messengers.
 - 2) Hormones are produced by endocrine glands in the body.
 - 3) Endorphins hormone are made up as steroids.
 - 4) Thyroxine produced in the thyroid gland is an iodinated derivative of amino acid tyrosine.
56. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion(A): Fructose give Seliwanoffs test faster than Glucose.
Reason (R): Fructose undergo β – elimination followed by formation of furfural.
 In the light of the above statements, choose the correct answer from the options given below:
- 1) **A** is false but **R** is true.
 - 2) Both **A** and **R** are true and **R** is the correct explanation of **A**.
 - 3) **A** is true but **R** is false.
 - 4) Both **A** and **R** are true but **R** is not the correct explanation of **A**.
57. For the preparation of meta nitro ethyl Benzene which is the best method

- 1)  $\xrightarrow[H_2SO_4]{HNO_3}$ (A) $\xrightarrow[Anhydrous AlCl_3]{CH_3COCl}$ (B) $\xrightarrow[KOH, \Delta]{NH_2-NH_2}$ (C)
- 2)  $\xrightarrow[Anhydrous AlCl_3]{CH_3COCl}$ (A) $\xrightarrow[H_2SO_4]{HNO_3}$ (B) $\xrightarrow[KOH, \Delta]{NH_2-NH_2}$ (C)
- 3)  $\xrightarrow[Anhydrous AlCl_3]{CH_3COCl}$ (A) $\xrightarrow[H_2SO_4]{HNO_3}$ (B) $\xrightarrow[HCl]{Zn(Hg)}$ (C)
- 4)  $\xrightarrow[H_2SO_4]{HNO_3}$ (A) $\xrightarrow[KOH, \Delta]{NH_2-NH_2}$ (B) $\xrightarrow[Anhydrous AlCl_3]{CH_3COCl}$ (C)





58. What Volume of $0.8M$ $AlCl_3$ solution should be mixed with 50 ml of $0.2M$ $CaCl_2$ solution to get a solution of chloride ion concentration equal to $0.6M$?
- 1) 5.56 ml 2) 100 ml 3) 50 ml 4) 4.89 ml

59. An increase in the rate of reaction with rise in temperature is due to

- 1) Decrease in the number of collisions.
 2) An increase in the number of activated molecules.
 3) Lowering of activation energy.
 4) Shortening of mean free path.

60. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$.

This is a gaseous phase reaction taking place in a $1L$ flask at 127° . Starting with 1 mole of N_2 and 3 mole of H_2 , the equilibrium mixture obtained is such that if it is liberated requires 500 ml of $1.0M$ H_2SO_4 for neutralization. Which of the following is the most appropriate K_c ?

- 1) 0.03 2) 0.59 3) 0.27 4) 0.22

61. The vapour pressure of pure X are given by the following equation:

$$\text{Solid X: } \ln P(\text{mmHg}) = 12 - \frac{3000(K)}{T}$$

$$\text{Liquid X: } \ln P(\text{mmHg}) = 6 - \frac{2400(K)}{T}$$

Find out the freezing point of X –

- 1) $100K$ 2) $120K$ 3) $150K$ 4) $200K$

62. The hydrolysis of an ester was carried out separately with $0.05M$ HCl and $0.025M$ H_2SO_4 $\left[k_{a1} = \infty \text{ but } K_{a2} = 1.2 \times 10^{-2} \right]$. Which of the following will be true, for the apparent rate constant.

- 1) $K_{HCl} > K_{H_2SO_4}$ 2) $K_{HCl} < K_{H_2SO_4}$
 3) $K_{HCl} = K_{H_2SO_4}$ 4) $K_{H_2SO_4} = 2K_{HCl}$

JEE MAIN
2023SINGARAJU
VENKAT KOUNDINYA
Sri Chaitanya
JEE 12th Class306
RANK

RANK

JEE Advanced
2023VAVILALA
CHIDVILAS REDDY
Sri Chaitanya
JEE 12th Class341
RANK

RANK

NEET
2023BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
JEE 12th Class28
RANK

RANK



63. The enthalpies of combustion of formaldehyde and paraformaldehyde (a polymer of formaldehyde) are -134 and -732 k Cal/mol respectively. The enthalpy of polymerization per mole of paraformaldehyde is -72k Cal. The molecular formula of paraformaldehyde is
- 1) $C_3H_6O_3$ 2) $C_5H_{10}O_5$ 3) $C_6H_{12}O_6$ 4) $C_7H_{14}O_7$
64. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is: $CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$.
- At 298K standard Gibb's energies of formation for $CH_3OH(l)$, $H_2O(l)$ and $CO_2(g)$ are -166.2, -237.2 and -394.4 $kJmol^{-1}$ respectively. If standard enthalpy of combustion of methanol is -826 $kJmol^{-1}$, efficiency of the fuel cell will be (approx.):
- 1) 80% 2) 85% 3) 90% 4) 97%
65. Choose the incorrect sequence of acidic nature of oxides ?
- 1) $SiO_2 < P_4O_{10} < SO_3 < Cl_2O_7$ 2) $CO_2 > SiO_2 > GeO_2 > SnO_2$
- 3) $Cl_2O > ClO_2 > Cl_2O_6 > Cl_2O_7$ 4) $MnO < Mn_2O_3 < MnO_2 < Mn_2O_7$
66. Identify the CORRECT match.

(I)	XeF_2	(A)	Central atom has sp^3 hybridisation and it has bent shape
(II)	N_3^-	(B)	Central atom has sp^3d^2 hybridisation and it has octahedral shape
(III)	PCl_6^-	(C)	Central atom has sp hybridization and it has linear shape
(IV)	ICl_2^+	(D)	Central atom has sp^3d hybridization and it has linear shape

- 1) I - A, II - B, III - C, IV - D 2) I - D, II - B, III - D, IV - C
- 3) I - B, II - C, III - A, IV - D 4) I - D, II - C, III - B, IV - A
67. Which of the following statements is incorrect?





- 1) Gallium (Ga) has highest boiling point and lowest melting point among group 13 elements
- 2) Among Ga, In, Tl and Al; Tl has highest $E_{M^{+3}/M}^0$ value (SRP)
- 3) Among group 13 elements, boron has the lowest density.
- 4) Aluminium chloride in acidified aqueous solution forms octahedral $[Al(H_2O)_6]^{3+}$ ion.
68. Which of the following statements is incorrect regarding 3d series d-block elements?
- 1) VCl_5 on hydrolysis gives $VOCl_3$ as one of the product
- 2) Highest fluorides of Mn and Cr are MnF_7 and CrF_6 respectively
- 3) The stability of $Cu_{(aq)}^{2+}$ rather than $Cu_{(aq)}^{+}$ is due to the much more negative $\Delta_{hyd}H^0$ of $Cu_{(aq)}^{2+}$ than $Cu_{(aq)}^{+}$. ($\Delta_{hyd}H^0$ = hydration enthalpy)
- 4) $VO_2^{+} < Cr_2O_7^{2-} < MnO_4^{-}$ is the correct order of oxidising power.
69. Consider the statements I to V and identify the correct statements:
- I. Both $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are diamagnetic complexes
- II. $[Ni(H_2O)_6]^{2+}$ is diamagnetic and $[Ni(NH_3)_6]^{2+}$ is paramagnetic ($Ni^{2+}:3d^8$).
- III. $[Cr(C_2O_4)_3]^{3-}$ has two geometrical isomers
- IV. Lower oxidation states of heavier members of group 6 of transition series are more stable.
- V. HCl is not used to make the medium acidic in oxidation reactions of $KMnO_4$ in acidic medium.
- 1) Only II 2) Only I and III 3) Only IV 4) Only V
70. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.





Assertion (A): Most of the reactions of fluorine are exothermic

Reason (R): Fluorine forms only one oxoacid while other halogens form a number of oxoacids.

In the light of the above statements, choose the correct answer from the options given below:

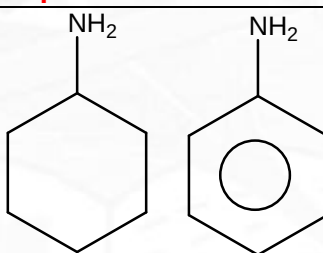
- 1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion
- 2) Assertion is true but Reason is false
- 3) Both Assertion and Reason are true and Reason is not the correct explanation for Assertion
- 4) Assertion is false but Reason is true

SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases



71. Find the number of tests by which can be distinguished.
- (i) Isocyanide test (ii) Hinsberg's test (iii) Mustard oil test
 (iv) Dye-azo test (v) Lucas test (vi) Victor Meyer's test
 (vii) Carbylamine test
72. If x is the number of electrons of $\text{Ca}(20)$ with $n + l + m = 3$. y is the number of radial nodes in orbital $4d_{z^2}$. w is the number of angular nodes in $4d_{z^2}$ orbital and q is the number of nodal planes in $3p_z$ orbital then, find the value of $[x + y + w + q]$?
73. A solution of $0.1\text{M } \text{CH}_3\text{COOH}$ is placed between parallel electrodes of cross-section area 4cm^2 , separated by 2cm . For this solution, resistance measured is 100Ω . Calculate





elevation in boiling point of the 0.1M CH_3COOH solution, using following informations

$$K_b = 0.8 K \cdot kg \cdot mol^{-1}; \Lambda_m^\infty(H^+) = 300 \text{ Scm}^2 \text{ mole}^{-1};$$

$$\Lambda_m^\infty(CH_3COO^-) = 100 \text{ Scm}^2 \text{ mole}^{-1}. \text{ Consider [Molarity = molality].}$$

Give your answer multiplying by 100.

74. Consider the elements: Si, Ge, C and Pb.

If 'M' is the element having lowest first ionisation enthalpy ($\Delta_i H_1$) value, then its stable oxidation state is '+x'

The value of 'x' is:

75. (i) 'x' is the number of planar molecules among SF_4, XeF_4, BrF_5, H_2O and SO_3 .

(ii) 'y' is the number of e_g electrons in $[Mn(CN)_6]^{3-}$ complex ion

(iii) 'z' is the number of unpaired electrons in $[Co(NO_2)_6]^{4-}$ complex ion

Then the value of (x + y + z) is:

**JEE MAIN****2023**SINGARAJU
VENKAT KOUNDINYA
Sri Chaitanya
JEE 12th Class**306****100****RANK****RANK****1****JEE Advanced**
2023VAVILALA
CHIDVILAS REDDY
Sri Chaitanya
JEE 12th Class**341****100****RANK****RANK****1****NEET****2023**BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
NEET 12th Class**200****100****RANK****RANK****1**



Sri Chaitanya
Educational Institutions

Infinity
Learn



Sri Chaitanya
Techno School

The right mentor for IIT (JEE), NEET, Olympiad & all Other Competitive exams



JEE MAIN 2025

31 STUDENTS BELOW 100 AIR

1 ALL INDIA OPEN CATEGORY RANK

300
300

VANGALA AJAY REDDY
APP. NO. 25030255592
CLASSROOM STUDENT FROM GRADE IX - X

1 ALL INDIA OPEN CATEGORY RANK

300
300

DEVPUTTA MAJHI
APP. NO. 250300016165
DLP/AITS STUDENT

9 ALL INDIA OPEN CATEGORY RANK

295
300

TOSHNIWAL SHIVEN
APP. NO. 25030391420
DLP/AITS STUDENT

10 ALL INDIA OPEN CATEGORY RANK

295
300

SAKSHAM JINDAL
APP. NO. 25030236696
DLP/AITS STUDENT

BELOW
100
ALL INDIA OPEN
CATEGORY RANKS

31

BELOW
500
ALL INDIA OPEN
CATEGORY RANKS

95

BELOW
10
ALL INDIA CATEGORY
RANKS COUNT

10

BELOW
100
ALL INDIA CATEGORY
RANKS COUNT

98

BELOW
1000
ALL INDIA CATEGORY
RANKS COUNT

579

**TOTAL QUALIFIED RANKS
FOR JEE ADVANCED-2025**

22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT
Sri Chaitanya Tops JEE ADVANCED

ALL INDIA OPEN CATEGORY RANKS

AIR

1

RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)

AIR

3

MAJID MUJAHID HUSAIN
H.T.No. 251134112*

AIR

5

UJJWAL KESARI
H.T.No. 252016104*

AIR

6

AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW
100
ALL INDIA OPEN
CATEGORY RANKS

29

BELOW
500
ALL INDIA OPEN
CATEGORY RANKS

113

BELOW
1000
ALL INDIA OPEN
CATEGORY RANKS

205

BELOW
1000
ALL INDIA CATEGORY
RANKS COUNT

745

**NUMBER OF
QUALIFIED RANKS**
4,212

*DLP/AITS

www.srichaitanya.net

040-66 06 06 06

For More Join: @IITJEE_Advanced